

## VYTEGRA, Russian Federation

WMO: 228370

Lat: 61.0168N

Lon: 36.4556E

Elev: 184

StdP: 14.60

Time Zone: 3.00 (E03)

Period: 90-00

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -22.3      | -15.1 | -26.8                         | 1.2 | -22.3 | -19.6 | 1.9 | -14.8 | 20.2                  | 24.0 | 17.3 | 25.9 | 0.9                   | 10   | 0.617 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.9                   | 81.4            | 66.7 | 76.8 | 63.7 | 73.5 | 62.1 | 67.7                | 77.8 | 65.8 | 74.7 | 63.7 | 71.1 | 6.0                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.1                            | 90.0 | 72.8 | 62.2 | 84.3 | 69.8 | 60.4 | 79.0 | 67.6 | 32.2          | 77.6 | 30.7 | 74.8 | 29.1 | 70.7 | 73.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 16.5              | 14.5 | 13.0 | -28.0                      | 87.0 | 6.0                | 4.0 | -32.3                                              | 89.9 | -35.8      | 92.2 | -39.1      | 94.5 | -43.5      | 97.4 |
|                   |      |      | -28.2                      | 70.9 | 5.9                | 2.5 | -32.4                                              | 72.7 | -35.9      | 74.1 | -39.2      | 75.5 | -43.6      | 77.3 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 37.9   | 16.0  | 16.6  | 26.5  | 37.8  | 46.3  | 58.3  | 61.9  | 57.8  | 48.5  | 38.4  | 25.2  | 19.7  |  |  |
|                                                                        | DBStd       | 19.00  | 13.59 | 15.45 | 8.90  | 8.96  | 8.99  | 7.91  | 6.25  | 5.71  | 7.66  | 8.50  | 11.12 | 12.96 |  |  |
|                                                                        | HDD50       | 5468   | 1054  | 936   | 729   | 380   | 181   | 12    | 1     | 8     | 118   | 366   | 745   | 938   |  |  |
|                                                                        | HDD65       | 9981   | 1519  | 1356  | 1194  | 817   | 581   | 233   | 135   | 229   | 496   | 823   | 1195  | 1403  |  |  |
|                                                                        | CDD50       | 1045   | 0     | 0     | 0     | 13    | 67    | 263   | 369   | 251   | 74    | 8     | 0     | 0     |  |  |
|                                                                        | CDD65       | 79     | 0     | 0     | 0     | 0     | 2     | 33    | 37    | 6     | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 700    | 0     | 0     | 0     | 2     | 31    | 337   | 282   | 41    | 7     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 165    | 0     | 0     | 0     | 0     | 6     | 96    | 63    | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 5.6    | 6.0   | 6.1   | 6.0   | 5.6   | 5.6   | 5.1   | 4.8   | 4.8   | 5.3   | 5.8   | 6.0   | 6.1   |  |  |
|                                                                        | PrecAvg     | 26.7   | 1.7   | 1.2   | 1.3   | 1.4   | 1.9   | 2.6   | 3.1   | 3.5   | 2.8   | 3.0   | 2.4   | 2.0   |  |  |
|                                                                        | PrecMax     | 37.2   | 2.6   | 3.0   | 3.2   | 3.7   | 4.0   | 5.8   | 8.9   | 8.8   | 5.7   | 6.7   | 4.7   | 4.4   |  |  |
| (15) Precipitation                                                     | PrecMin     | 19.4   | 0.4   | 0.2   | 0.1   | 0.2   | 0.1   | 0.3   | 0.7   | 0.6   | 0.6   | 0.0   | 0.6   | 0.6   |  |  |
|                                                                        | PrecStd     | 3.5    | 0.6   | 0.6   | 0.6   | 0.7   | 0.8   | 1.2   | 1.7   | 1.6   | 1.1   | 1.4   | 0.9   | 0.8   |  |  |
|                                                                        | DB          | 37.0   | 41.2  | 49.3  | 70.6  | 77.6  | 88.0  | 85.9  | 78.2  | 74.6  | 62.2  | 46.5  | 39.1  |       |  |  |
|                                                                        | MCWB        | 34.8   | 37.9  | 42.2  | 54.0  | 61.2  | 70.0  | 67.7  | 63.7  | 63.7  | 56.7  | 44.4  | 37.5  |       |  |  |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2% DB       | 35.7   | 38.4  | 42.6  | 63.9  | 71.2  | 82.6  | 81.2  | 74.5  | 68.1  | 56.2  | 42.8  | 37.3  |       |  |  |
|                                                                        | 2% MCWB     | 33.9   | 36.2  | 36.8  | 51.2  | 57.3  | 67.8  | 66.5  | 62.1  | 60.5  | 51.7  | 41.1  | 35.9  |       |  |  |
|                                                                        | 5% DB       | 34.4   | 35.9  | 39.8  | 57.7  | 66.6  | 77.6  | 76.5  | 71.6  | 63.0  | 52.2  | 40.6  | 35.9  |       |  |  |
|                                                                        | 5% MCWB     | 32.8   | 33.9  | 35.5  | 47.4  | 55.0  | 64.6  | 63.9  | 61.8  | 57.0  | 48.6  | 39.0  | 34.3  |       |  |  |
|                                                                        | 10% DB      | 33.1   | 34.2  | 37.6  | 52.2  | 62.1  | 73.2  | 73.3  | 68.6  | 59.1  | 49.7  | 37.8  | 34.4  |       |  |  |
|                                                                        | 10% MCWB    | 31.7   | 32.6  | 34.3  | 44.4  | 52.8  | 61.7  | 62.7  | 60.2  | 54.8  | 47.3  | 36.6  | 32.8  |       |  |  |
|                                                                        | 0.4% WB     | 35.1   | 38.5  | 42.2  | 55.0  | 63.9  | 71.1  | 70.0  | 66.6  | 65.0  | 57.1  | 44.3  | 37.9  |       |  |  |
|                                                                        | 0.4% MCDB   | 36.1   | 40.4  | 47.8  | 68.9  | 75.3  | 84.2  | 81.3  | 73.9  | 72.0  | 62.7  | 45.7  | 38.6  |       |  |  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 34.1   | 36.5  | 38.3  | 51.8  | 59.8  | 68.6  | 67.6  | 64.3  | 61.0  | 52.1  | 41.6  | 36.1  |       |  |  |
|                                                                        | 2% MCDB     | 35.2   | 38.2  | 41.1  | 62.6  | 67.4  | 80.0  | 77.3  | 70.9  | 66.9  | 55.5  | 42.8  | 37.1  |       |  |  |
|                                                                        | 5% WB       | 33.0   | 34.1  | 36.3  | 48.3  | 56.1  | 66.0  | 65.9  | 62.8  | 57.9  | 49.3  | 39.2  | 34.6  |       |  |  |
|                                                                        | 5% MCDB     | 34.2   | 35.6  | 38.9  | 56.2  | 65.4  | 75.9  | 74.9  | 68.9  | 62.2  | 51.6  | 40.7  | 35.7  |       |  |  |
|                                                                        | 10% WB      | 31.8   | 32.7  | 34.6  | 45.0  | 53.0  | 63.4  | 63.7  | 61.3  | 55.2  | 47.2  | 36.8  | 33.1  |       |  |  |
|                                                                        | 10% MCDB    | 33.2   | 34.3  | 37.0  | 51.7  | 61.9  | 70.7  | 71.3  | 67.4  | 59.2  | 49.7  | 37.9  | 34.3  |       |  |  |
|                                                                        | MDBR        | 11.8   | 12.8  | 13.8  | 15.1  | 15.9  | 17.1  | 15.9  | 13.8  | 11.8  | 8.8   | 8.5   | 9.8   |       |  |  |
|                                                                        | 5% DB       | 7.9    | 8.4   | 13.8  | 24.0  | 23.6  | 23.3  | 22.1  | 19.6  | 16.8  | 12.2  | 7.1   | 6.0   |       |  |  |
| (36) Mean Daily Temperature Range                                      | MCWBR       | 7.9    | 7.3   | 9.7   | 13.8  | 12.9  | 11.7  | 9.9   | 10.7  | 9.9   | 8.5   | 6.5   | 5.8   |       |  |  |
|                                                                        | MCDBR       | 7.6    | 8.4   | 9.6   | 22.6  | 21.8  | 21.0  | 18.8  | 16.5  | 15.0  | 11.0  | 6.1   | 6.5   |       |  |  |
|                                                                        | MCWBR       | 7.7    | 7.7   | 7.4   | 13.4  | 12.6  | 11.3  | 8.8   | 10.4  | 9.3   | 8.0   | 5.9   | 6.4   |       |  |  |
|                                                                        | MCDBR       | 7.7    | 7.7   | 7.4   | 13.4  | 12.6  | 11.3  | 8.8   | 10.4  | 9.3   | 8.0   | 5.9   | 6.4   |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.256  | 0.303 | 0.327 | 0.368 | 0.358 | 0.354 | 0.391 | 0.377 | 0.344 | 0.325 | 0.279 | 0.228 |       |  |  |
|                                                                        | taud        | 2.036  | 2.074 | 2.130 | 2.259 | 2.433 | 2.469 | 2.385 | 2.403 | 2.478 | 2.383 | 2.132 | 1.984 |       |  |  |
|                                                                        | Ebn at Noon | 142    | 206   | 246   | 260   | 275   | 278   | 262   | 256   | 246   | 206   | 134   | 99    |       |  |  |
|                                                                        | Edh at Noon | 13     | 24    | 32    | 35    | 32    | 31    | 33    | 30    | 23    | 18    | 12    | 9     |       |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 83     | 275   | 705   | 1198  | 1593  | 1687  | 1623  | 1235  | 705   | 282   | 81    | 42    |       |  |  |
|                                                                        | RadStd      | 11     | 69    | 96    | 133   | 104   | 107   | 196   | 94    | 69    | 40    | 14    | 6     |       |  |  |

## Historical Trends

|                        |  | Heating |        | Cooling |       |       | Degree-Days |       |       |       |     |
|------------------------|--|---------|--------|---------|-------|-------|-------------|-------|-------|-------|-----|
|                        |  | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |     |
| Station Only           |  | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   | (m) |
|                        |  | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   | N/A |
| Regional (5 neighbors) |  | +1.03   | N/A    | N/A     | N/A   | N/A   | N/A         | -394  | -418  | N/A   | N/A |

Nomenclature: See separate page